

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

8875/01

Paper 1 Multiple Choice

24 September 2009

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

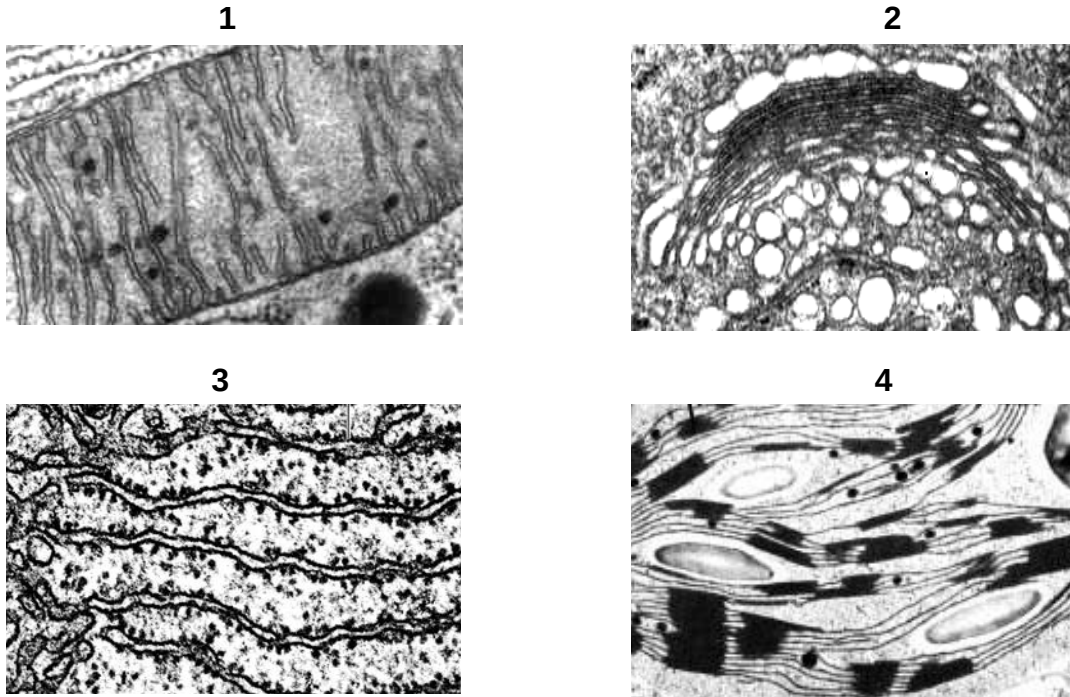
Any rough working should be done in this booklet.

Calculators may be used.

This document consists of **12** printed pages.



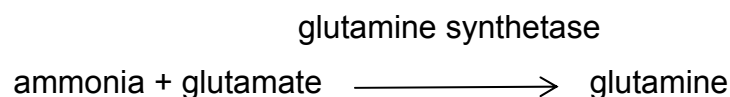
- 1 The following contains images of a number of structures that can be seen in a cell.



Identify structures **1 - 4** and the process that occurs in structure **4**.

	1	2	3	4	Process
A	chloroplast	nucleus	ribosome	mitochondrion	respiration
B	mitochondrion	Golgi apparatus	rough ER	chloroplast	photosynthesis
C	mitochondrion	Golgi apparatus	smooth ER	rough ER	protein synthesis
D	chloroplast	nucleolus	Golgi apparatus	microfilaments	translocation

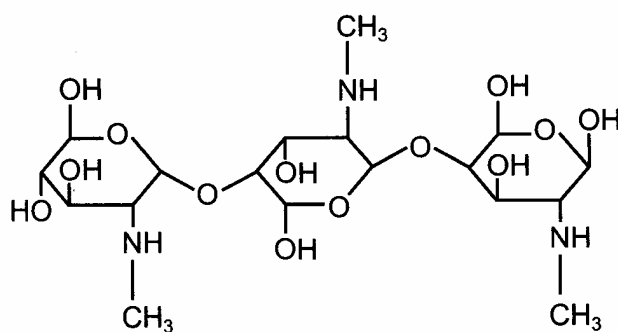
- 2 The diagram shows how the enzyme glutamine synthetase removes the ammonia produced during plant metabolism.



Some herbicides contain an active agent which resembles glutamate. What is the likely mode of action of this agent?

- A** It acts as an end-product inhibitor.
- B** It acts as a competitive inhibitor.
- C** It decreases levels of ammonia.
- D** It increases levels of glutamate.

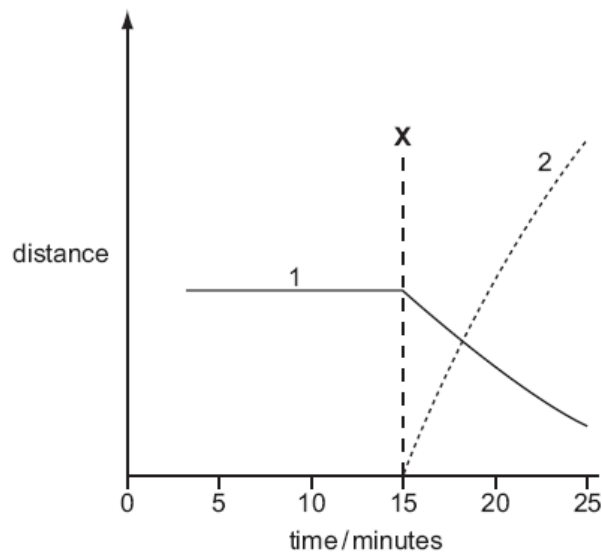
- 3 The diagram shows the structure of a competitive inhibitor of the enzyme lysozyme.



Which substance is most likely to be the normal substrate of lysozyme?

- A Lipid
 - B Polynucleotide
 - C Polypeptide
 - D Polysaccharide
- 4 Which of the followings is are / true about cellulose and collagen?
- 1 Both serve as structural components as they have high tensile strength.
 - 2 The high tensile strength is a result of interchain hydrogen bonds.
 - 3 Both show quaternary level of organization.
 - 4 Cellulose molecules can only form hydrogen bonds whereas collagen is capable of forming other bond in addition to hydrogen bonds.
- A 1 only
 - B 1 and 2 only
 - C 1, 2 and 3 only
 - D All of the above

- 5 The graph shows measurements during one mitotic cell cycle. Which stage of mitosis commences at X and which measurements are illustrated by curves 1 and 2?



	Stage beginning at X	Distance between centromeres of chromosomes and poles of spindle	Distance between the centromeres of sister chromatids
A	Anaphase	1	2
B	Anaphase	2	1
C	Metaphase	1	2
D	Metaphase	2	1

- 6 What is the basis for the difference in the synthesis of the leading and lagging strands of DNA molecules?
- I The anti-parallel arrangement of the DNA strands.
 - II The RNA primers are required to initiate DNA elongation.
 - III DNA polymerase can join new nucleotides to the 3' end of the growing strand.
 - IV Helicase and single-stranded binding proteins work at the 5' end of the DNA strand.
- A** II and IV
B I and III
C I and IV
D II and III

- 7 Which sugar and base, in addition to inorganic phosphate, will be released from the hydrolysis of a certain pyrimidine nucleotide?

<i>sugar</i>	<i>base</i>
A deoxyribose	uracil
B deoxyribose	guanine
C pentose	uracil
D ribose	adenine

- 8 Cytosine comprises 0.32 of the nitrogenous bases in a sample of DNA consisting of 200 base pairs.

What is the total number of adenine and thymine bases in a single strand of DNA from this sample?

- A 16
B 18
C 36
D 72

- 9 The table shows DNA triplets and their corresponding amino acids.

DNA triplet	AGG	ATG	CGC	CGT	GCA	GCG	TAC	TCC	TGG
amino acid	ser	tyr	ala	ala	arg	arg	met	arg	thr

Part of the mRNA sequence produced following transcription of a gene is shown.

GCAGCGUACUCC

What is the correct sequence of amino acids produced when this mRNA is translated?

- A ala-ala-thr-ser
B ala-ala-tyr-ser
C arg-arg-met-arg
D arg-arg-met-thr

- 10 Which of the following shows the order in which the structures would be used in the process of gene expression resulting in the formation of a protein?

<i>earliest</i>	→			<i>latest</i>
A polyribosome	promoter	spliceosome	primary transcript	
B spliceosome	promoter	primary transcript	polyribosome	
C promoter	primary transcript	spliceosome	polyribosome	
D primary transcript	spliceosome	promoter	polyribosome	

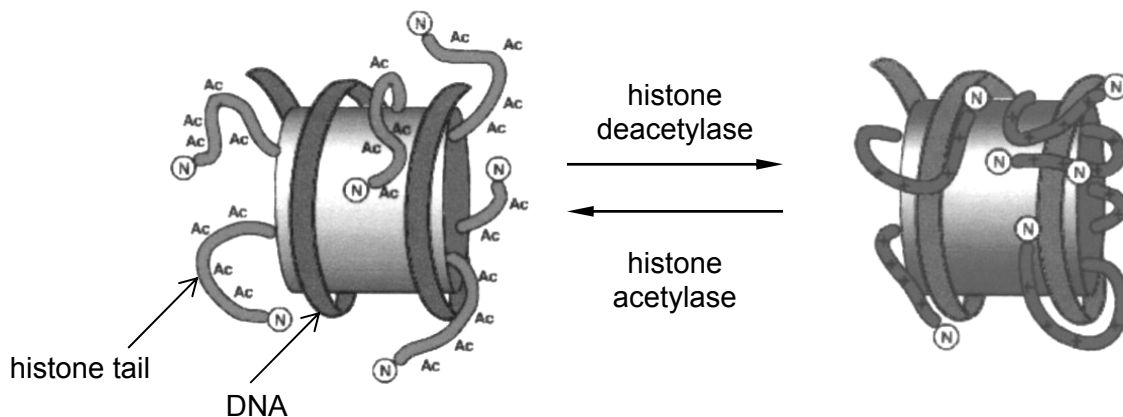
11 Which of the following **does not** describe the function of telomeres?

- A maintain integrity of genes
- B prevent cancerous cell growth
- C prevent chromosome fusion
- D regulate gene transcription

12 Which of the following is **not** a form of gene regulation in eukaryotes?

- A chromosome modification
- B inducible operons
- C interaction of control elements
- D RNA splicing

13 The diagram below shows how acetylation of histones promotes loose chromatin structure. Recent evidence has shown that chemical modification of histones play a direct role in regulation of gene expression. Which of the following best explains how acetylation regulates gene expression?



- A Helicase action is enhanced by acetylation.
- B Acetylation of histones neutralizes their negative charges and encourages binding to DNA polymerase.
- C When nucleosomes are highly acetylated, chromatin becomes less compact and DNA is more accessible for transcription.
- D RNA polymerase works better by binding with acetyl groups.

- 14** In a small mammal, the allele for grey fur, **G**, is dominant to that for white fur, **g**. The allele for long tail, **T**, is dominant to the allele for short tail, **t**. Animals with grey fur and long tails were crossed with those having white fur and short tails.

The table shows the phenotypes of the 55 offspring.

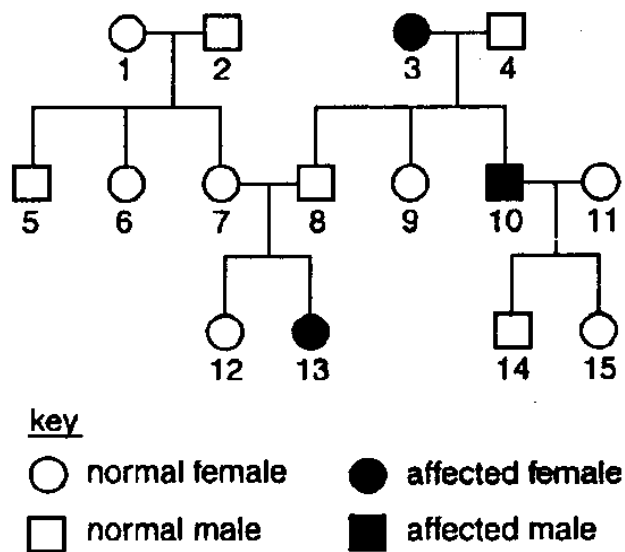
<i>Number of offspring</i>	<i>fur</i>	<i>tail</i>
15	grey	long
14	grey	short
14	white	long
12	white	short

What were the genotypes of the parents?

- A** Gg^tt x gg^tt
B GG^Tt x Gg^tt
C Gg^Tt x Gg^Tt
D Gg^Tt x gg^tt

Refer to the following family tree for Questions **15** and **16**.

- 15** The family tree shows the inheritance of a genetic condition governed by the R/r gene.



Which of the females are certain to have the genotype Rr?

- A** 1, 6 and 7
B 1, 7 and 12
C 7, 9 and 15
D 9, 12 and 15

16 What is the probability of **13** and **14** having an affected daughter?

- A** 0
- B** 0.25
- C** 0.5
- D** 0.75

17 In maize plants, two gene loci on different chromosomes have the following alleles.

Gene locus one
R : green stems
r : red stems

Gene locus two
D : normal height
d : dwarf plants

The following cross was carried out and the seeds collected.

Rr Dd x **Rr Dd**

1600 of the seeds were grown to examine the phenotypes of the offspring.

It would be reasonable to expect that among the offspring, approximately

- A** 100 offspring would be homozygous at both gene loci.
- B** 600 of the offspring will have green stems and be of normal height.
- C** $\frac{1}{3}$ of the red stem, normal height offspring will be heterozygous at both gene loci.
- D** $\frac{2}{3}$ of the green stem, dwarf offspring will be heterozygous at gene locus one

18 A man with **O** blood type married a woman with **AB** blood type and they have four children in the family. The first has **A** blood type, the second has **B** blood type, the third has **AB** blood type and the fourth has **O** blood type.

Which of the following conclusions is possible from the information given?

- A** The first two children are adopted as they do not share the blood type of their parents
- B** The third and fourth children are probably adopted as the parents cannot produce children with these phenotypes
- C** Only the third child belonged to the parents and the others are probably adopted as the **A** and **B** allele are codominant
- D** It is possible that all four children are the offspring of the pair of parents because of independent assortment of chromosomes

- 19 It has been found that an aqueous suspension of isolated chloroplasts will evolve oxygen if illuminated in the presence of a certain type of compound.

Which type of compound and which wavelength of light would give maximum oxygen evolution?

<i>type of compound</i>	<i>wavelength of light at which maximum evolution occurs / nm</i>
A electron acceptor	300 and 550
B electron acceptor	450 and 700
C electron donor	680 and 700
D electron donor	420 and 600

- 20 Which of the following is **true** about Calvin cycle?

- A Its cessation will cause light dependent reactions to stop.
- B It can only take place under dark conditions.
- C It occurs in the thylakoid space.
- D Its progression is independent of the light dependent reactions.

- 21 Which statement about respiration is **true**?

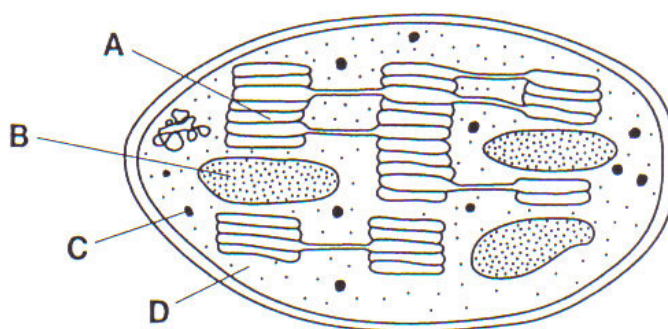
- A Cell respiration can only take place in mitochondria.
- B CO_2 is released in the conversion of glucose to pyruvate.
- C Glycolysis takes place in the mitochondrion.
- D Oxygen is the terminal electron acceptor in aerobic respiration.

- 22 Some organisms change the composition of their membranes to maintain membrane fluidity when temperature changes.

Which changes occur in the composition of membranes following a change to low temperature?

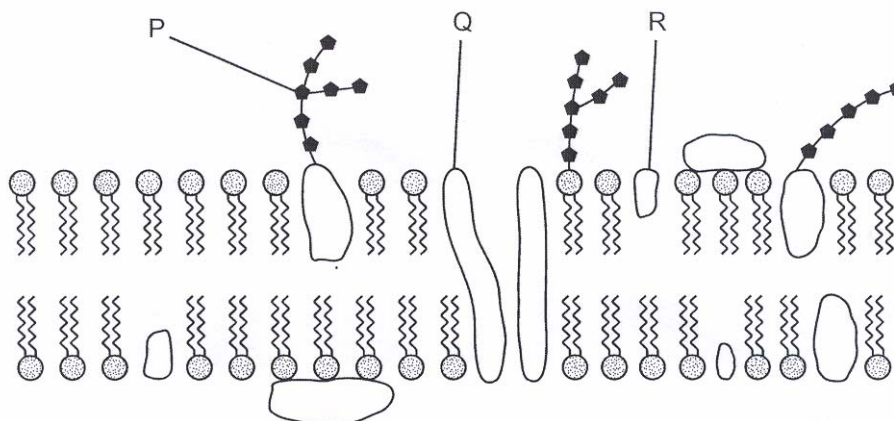
- A Increase phospholipids, decrease cholesterol
- B Increase saturated fatty acids, increase cholesterol
- C Increase unsaturated fatty acids, increase cholesterol
- D Increase saturated fatty acids, decrease cholesterol

- 23 The diagram represents a section through a chloroplast.



In which part does chemiosmosis takes place?

- 24 The diagram below shows part of a cell surface membrane.



What is the correct function for each of the structures labelled?

	regulates membrane fluidity	forms hydrogen bonds with water to stabilise membrane	transports ions and large polar molecules
A	R	R	Q
B	P	Q	R
C	Q	R	P
D	R	P	Q

- 25 Which of the following ideas were **not** parts of Darwin's theory of evolution by natural selection?

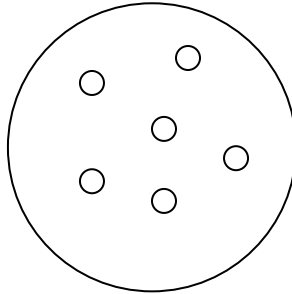
- I Acquired variation is inherited
- II Variation between individuals arises by gene mutation
- III Organisms produce more offspring than the environment can support
- IV Only those individuals best adapted to the environment survive and reproduce
- V Regions that encode portions of the polypeptide that are vital for structure and function are less likely to incur mutations

- A** I, II and V
- B** I, III and IV
- C** II, III and V
- D** II, IV and V

- 26 A large population of equal numbers of dark and light mice was released into an area where owls are predators of mice. Because of predation, only 25% of these mice survived and selective killing by owls changed the proportions of dark to light mice from 1:1 to 4:1.

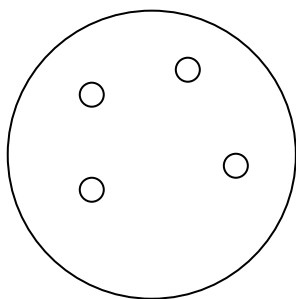
If mice produce an average of 8 offspring per litter and the pattern of predation remains the same, what would happen to the population of light mice?

- A They disappear after one further generation.
 - B They disappear after two further generations.
 - C They remain at the level of one fifth of the population.
 - D They would be reduced to a constant but very low frequency.
- 27 A mixture of insulin gene, plasmid with ampicillin and tetracycline resistant genes, *EcoRI* and DNA ligase was allowed to react and subsequently heat shocked with *E. coli*. The mixture was then plated on nutrient agar to obtain a master plate with colonies shown below.

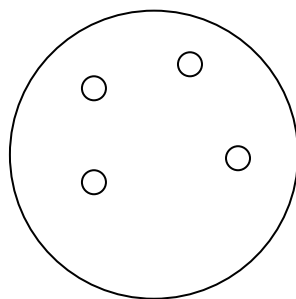


Master plate (nutrient agar)

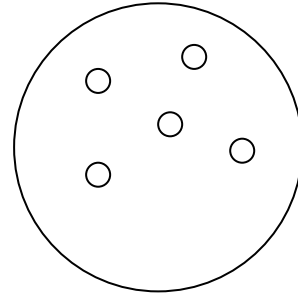
Three replica plates containing different antibiotics were obtained from the master plate. Diagram below shows the results of the replica plates.



Nutrient agar
with ampicillin



Nutrient agar with
ampicillin and tetracycline



Nutrient agar
with tetracycline

Which of the following can be concluded from the results obtained?

- A There is one untransformed colony.
- B There are two colonies transformed with recombinant plasmid.
- C The ampicillin resistant gene is inactivated by insertion of the insulin gene.
- D The tetracycline resistant gene does not act as a selectable marker.

- 28 Which of the following would *not* be true of the polymerase chain reaction?
- A It could be used to amplify a specific target segment of DNA
 - B It consists of a three-step cycle involving DNA denaturation, annealing and extension.
 - C It has an advantage over cell-based cloning due to its sensitivity and accuracy.
 - D It requires a heat-stable DNA polymerase to extend the primers.
- 29 Where in the human body do stem cells giving rise to lymphocytes divide?
- A blood stream
 - B bone marrow
 - C liver
 - D lymph nodes
- 30 Maize varieties are being developed in which the leaves produce proteins that are toxic to insects. The DNA coding for these toxic proteins was inserted into a maize chromosome via a bacterial plasmid. Many people are opposed to this process.

Which objection is **not** biologically valid?

- A Beneficial insects may be killed if they eat genetically modified maize.
- B Genes for antibiotic resistance are present in plasmids and these genes may pass to harmful bacteria.
- C Hybridisation may transfer the bacterial genes from maize to weeds, giving the weed species new and harmful characteristics.
- D Mutations may be caused in cattle or humans that eat the genetically modified maize.